

RESEARCH ARTICLE

Green Innovation and Low-Carbon Operations: The Role of Net-Zero Capabilities in Transition Towards Carbon Neutrality in Circular Supply Chains

Usama Awan¹  | Saqib Shamim^{2,3} | Lea Hannola⁴

¹Center for Research on Digitalization and Sustainability, University of Inland Norway, Lillehammer, Norway | ²School of Business and Management, Queen Mary University of London, London, UK | ³Innolab, University of Vaasa, Vaasa, Finland | ⁴Department of Industrial Engineering and Management, Lappeenranta-Lahti University of Technology, Lappeenranta, Finland

Correspondence: Usama Awan (awan.usama@gmail.com)

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ABSTRACT

Reducing carbon emissions and achieving net-zero goals remain critical global supply chain priorities, and the transition to a circular supply chain is pivotal in reducing carbon emissions. This study empirically examines how low-carbon operations management practices impact circular supply chain management. We collected and analysed data from 183 respondents. Results from partial least squares structural equation modelling show that net-zero capabilities have a direct positive moderating effect on green product innovation and enhance end-of-product-life management and that net-zero capability can play a major role as a boundary-spanning function in accelerating green product innovation while balancing carbon emissions in the circular supply chain. Net-zero capability refers to a firm's ability to reduce its greenhouse gas emissions and align its operations with decarbonisation efforts. For practitioners, our results highlight that firms should focus on more than just green innovation; they should strengthen their net-zero capabilities to support sustainable supply chain practices.

1 | Introduction

Understanding the interdependence of low-carbon operations in mitigating environmental degradation and carbon emissions in production is a fundamental issue for many firms (de Sousa Jabbour et al. 2019; Wanke et al. 2021). The transition towards net-zero carbon emissions (reducing CO₂ released from energy and industrial activities) has become a growing concern due to challenges and barriers in the supply chain (Dohale et al. 2024; Okorie et al. 2023). It is necessary to find ways to integrate regenerative or restorative production systems (Chen et al. 2020; De Lima and Seuring 2023; Farooque, Zhang, Thürer, et al. 2019). Low-carbon emissions are emerging as a managerial philosophy through which firms can achieve their net-zero and sustainability objectives (Deng et al. 2024) by improving resource efficiency and end-of-product-life management (Abdul-Rashid et al. 2017;

Guo and Tsinopoulos 2024). Manufacturing firms face challenges regarding how to implement low-carbon operations management and its actual effects on circular supply chains (Lee and Hussain 2024).

Farooque, Zhang, and Liu (2019) conceptualised 'circular supply chain' (CSC) management as 'restorative and regenerative processes, appropriate business models (closed and open loop) and supply chain functions (reorientation) to achieve a zero-waste vision' (p. 896). CSC management effectively protects the natural ecosystem, progressing towards complete resource recovery to tackle material disruptions (Farooque, Zhang, and Liu 2019; Lahane et al. 2023). Previous research has reported the benefits of implementing low-carbon emissions practices for CSCs (Fu et al. 2023), but research on what capabilities and innovative strategies could contribute to this implementation is lacking

(Farooque et al. 2022; Li et al. 2023). Research on issues surrounding CSCs and low carbon emissions is of interest to both practitioners and academicians (S. Wang et al. 2024; Wanke et al. 2021). The idea of balancing the emission and elimination of greenhouse gas (GHG) in the environment, leaving zero carbon dioxide, is known as 'net zero' (Okorie et al. 2023; Vimal et al. 2022). Implementing net zero is challenging for firms (Kannan et al. 2022).

A recent study on mitigating climate change and accelerating carbon performance initiatives highlighted that success relies on a firm's ability to leverage its resources effectively (Morrison et al. 2025). However, an understanding of what encourages a firm to be involved in low-carbon operations and implement systems for material recovery is lacking (Fu et al. 2023; Lee and Hussain 2024; Lopes de Sousa Jabbour et al. 2021; Zhang et al. 2023). Tang et al. (2023) emphasised the importance of green innovation, which contributes to resource conservation and enables firms to reduce GHG emissions to expedite the transition towards a sustainable and carbon-neutral future. Empirical evidence suggests that greenhouse gas management processes, along with the integration of sustainability-related governance, may improve the reduction of GHG emissions (Morrison et al. 2024). A gap exists in the literature regarding the potential of net-zero emissions and carbon neutrality in supply chain management (Chen et al. 2024). Net-zero capabilities include a wide range of practices aiming to accelerate decarbonisation in the supply chain (Balci and Ali 2024). There is little empirical evidence of actual low-carbon operations' impact on the CSC (Lopes de Sousa Jabbour et al. 2021).

In developing countries, 'low-carbon operations management' is becoming increasingly concentrated, and institutions encourage firms to integrate circular material flows to tackle the challenges of material dependence on finite resources (de Sousa Jabbour et al. 2019). Böttcher and Müller (2015) defined low-carbon operations as 'integrating carbon efficiency into the planning, execution, and control of business processes to obtain competitive advantage' (p. 497). Low-carbon operations management is usually implemented throughout the product life for design and development (Lee and Hussain 2024; S. Wang et al. 2024).

Many scholars have sought to better understand why innovation processes are important in facilitating firm competitiveness (e.g., Hu et al. 2022). A defining characteristic of green innovation is that firms can increase productivity and minimise resource consumption (Chen 2008; Song et al. 2020). Recently, research has highlighted the role of green product innovation (GPI) in achieving firm performance (Negi et al. 2023). GPI involves developing new products, services or technologies with reduced environmental impact (Chen 2008). GPI practices aim to improve energy efficiency, durability and recyclability to minimise waste generation and reduce resource consumption (Bag et al. 2022; Chai et al. 2024). By implementing low-carbon processes, firms can decrease waste generation and design energy-efficient products (Bui et al. 2023). However, gaps remain in the research on net-zero emissions across various sectors, and addressing this gap could enrich academic knowledge and 'contribute to a deeper understanding and more effective implementation of net-zero strategies in supply chains' (Chen et al. 2024, 19).

Sandberg (2023) highlighted that to reduce emissions in the supply chain, firms require varied capabilities to implement CSCs effectively (Chari et al. 2022; Lu et al. 2024). For example, in a footwear industry incident in China, the export of shoes was halted due to non-compliance with a low-carbon process for the glue used in manufacturing (Asia Monitor Resource Centre 1997). To minimise their carbon emission footprints, IKEA implemented stricter carbon emission standards in 2014, and large corporations such as Disney and Microsoft have introduced internal carbon taxes to encourage emissions reductions (Sun et al. 2020). Adidas, a market leader in the sports retailer market, announced it was prioritising suppliers and manufacturers that use recyclable material throughout the entire value chain to achieve carbon neutrality by 2050 (Adidas 2023). This suggests that firms should recognise the importance of internal practices aimed at resource optimisation and reducing carbon emissions.

There is a lack of studies addressing the benefits of net-zero capabilities (Balci and Ali 2024). More research studies are needed to understand the mechanisms through which net-zero capabilities can optimise resource usage and transform manufacturing processes to achieve low carbon emissions. The purpose of this study is to examine how the implementation of low-carbon operations management (LCOM) practices influences closed-loop supply chains and 'end-of-life product management' through the mediating role of GPI within manufacturing firms. It also investigates how net-zero capabilities moderate the relationship between GPI initiatives and the advancement of closed-loop supply chains and end-of-life product management. Two research questions guided the study. How does implementing low-carbon operations management practices impact the closed-loop supply chain and end-of-life product management through GPI within manufacturing firms? How does the level of net-zero capability influence the contribution of GPI initiatives to the advancement of the closed-loop supply chain and end-of-life product management?

The manufacturing industry was selected because of its significance in global consumption, production and waste generation. This study makes a number of contributions. First, we contribute to the CSC management literature by examining the antecedents of LCOM in engineering manufacturing firms. Several prior research papers have empirically investigated the management of supply chain collaboration at the business level and its impact on performance results (Amir et al. 2023; Farooque et al. 2022; Li et al. 2023).

Second, prior studies have emphasised the importance of LCOM as a critical supply chain strategy for moving towards a future of complete resource recovery (Li et al. 2023; Sheng et al. 2023). This study explains the importance of low-carbon operations for GPI and improving CSC management practices. Third, this study is novel as it explores the moderating role of net-zero capability (Balci and Ali 2024), viewing it through the dynamic capabilities (DCs) perspective and explaining how firms create value by developing organisational processes (Augier and Teece 2009; Teece 2018). According to Balci and Ali (2024), net-zero capability also involves monitoring and setting ambitious targets, tracking progress and continuously improving sustainability material performance to achieve and maintain carbon neutrality. This

study empirically shows that net-zero capability enables firms to optimise material resources and minimise carbon emissions, aiding the shift towards material circularity in the supply chain.

2 | Literature Review

2.1 | Dynamic Capability and Net-Zero Capability

Dynamic and net-zero capabilities have attracted considerable attention in the transition towards carbon neutrality in the circular manufacturing supply chain (Chari et al. 2022). The dynamic capability (DC) view suggests that firms evolve and maintain a sustainable competitive advantage by leveraging capabilities (sensing, seizing and integrating) to protect and re-configure organisational capabilities and resources (Teece 2018, 2014). DCs are routines that enable organisations to coordinate and redeploy practices effectively (Augier and Teece 2007). They are essential in driving sustainability initiatives, helping companies recognise and embrace opportunities to achieve zero carbon contributions (Lu et al. 2024). To successfully plan and execute circularity efforts, the DC perspective lays out the steps and framework for making the most of available resources, for example, (de Sousa Jabbour et al. 2019; Farooque et al. 2022). In an uncertain environment, DCs enable firms to thrive and implement sustainability measures across production processes to attain net-zero targets (Okorie et al. 2023). Net-zero capabilities empower manufacturing firms by helping them sense opportunities for material efficiency, seize processes for prioritising energy efficiency, and implement sustainable and environmentally friendly practices to produce goods with a lower carbon footprint (Balci and Ali 2024). Based on the literature, we refer to net-zero capabilities as involving plans to reduce decarbonisation throughout the product life cycle, adapt to climate neutrality and promote a regenerative economy.

Net-zero capabilities include a wide range of practices that accelerate decarbonisation in the supply chain (Balci and Ali 2024). Decarbonisation in the supply chain may involve implementing systems to promote material efficiency and manage products' end-of-life processes, ultimately reducing the need for virgin materials (de Sousa Jabbour et al. 2019). The aim of net-zero capabilities is accelerated decarbonisation in the organisation's internal resources and capabilities (Balci and Ali 2024; Bhatia et al. 2022). The term net-zero capabilities can be interpreted as a firm's climate-neutral operational and technological capabilities that aim to reduce greenhouse gas emissions. In the context of operations management, these help us define net zero capability as follows: we therefore refer to net-zero capability as a firm's ability to reduce its greenhouse gas emissions and align its operations with decarbonisation efforts.

Recent research has recognised the role of LCOM in enhancing material efficiency, environmentally friendly design and green production processes (De Angelis et al. 2018; Lee and Hussain 2024; Lopes de Sousa Jabbour et al. 2021). For instance, Apple has worked with its manufacturing partners to achieve carbon neutrality by 2030, tackling carbon emissions through technological improvements in processes and materials and innovative carbon removal solutions (Apple 2020). According to Adu et al. (2023), reducing carbon emissions means firms must

increase green innovation to reduce pollution and maintain ecological balance.

Improving material efficiency and minimising waste generation in manufacturing processes is essential for CSC management (Lee and Hussain 2024). Batista et al. (2018) defined CSC management as 'the coordinated forward and reverse supply chains via purposeful business ecosystem integration for value creation from products or services, by-products, and useful waste flows through prolonged life cycles that improve the economic, social and environmental sustainability of organizations' (p. 446). CSC management entails the integration of key circular initiatives, such as prolonging the life cycle of products, reusing, recycling and remanufacturing (Farooque, Zhang, and Liu 2019; Farooque, Zhang, Thüerer, et al. 2019; Zhang et al. 2023). CSC management encourages the use of modular manufacturing systems to increase efficiency and decrease waste (Lahane et al. 2023; Mangla et al. 2018). The objective of circular production systems is to keep materials continuously circulated and reused within the production process and implement circular measures (waste reduction, lean manufacturing and material recovery) to achieve cost efficiency and promote the transition towards net zero (Batista et al. 2018; Govindan and Hasanagic 2018).

Sheng et al. (2023) empirically examined the relationship between LCOM and digital transformation for firm performance and found that low-carbon operations can transform production and manufacturing processes to align with circularity principles. Li et al. (2023) investigated the relationship between eco-innovation and CSC management; their conceptual framework includes public policy measures as an antecedent of CSC management. They argued that organisation efforts must track and monitor upstream emissions to implement CSC management successfully.

2.2 | Impact of Low-Carbon Operations on Green Innovation

Engineering manufacturing firms in developing countries often have limited access to clean technologies, and as a result, industrial activities have become a growing concern in efforts to reduce their carbon footprints (Sun et al. 2020). Engineering and technology management scholars recognise that low-carbon technologies may be fundamental drivers of long-term carbon emission reduction and resource efficiency (Chen et al. 2021). Due to low-carbon technology innovation, firms are investing in their production processes, indicating a potential link between low-carbon operations and a sustainable supply chain. Recent research has conceptualised technology innovation, bringing green innovation to resource-based industries (Azam et al. 2023) and increasing emission reduction incentives. A closed-loop system includes diverse practices and strategies to optimise resources and promote recycled material (Geng et al. 2021; Reimann et al. 2019). Circular production emphasises closed-loop manufacturing systems in which materials are continuously circulated and reused (Govindan et al. 2015; Mishra et al. 2023; Östlin et al. 2008). This highlights the importance of exploring further the role of low-carbon operations and how they impact source efficiency (Chen et al. 2021).

Some studies have shown that low-carbon products connect and integrate product life extension strategies (repair, refurbishment and remanufacturing) to reduce the need for new material to be extracted (Kumar et al. 2023; S. Wang et al. 2024). However, the role of LCOM in change towards net zero has largely been neglected (Kannan et al. 2022; Kumar et al. 2023). LCOM includes different activities to reduce carbon emissions in operations and to manage a successful transition towards achieving circularity in the material (Shaharudin et al. 2019). Recent research has emphasised the need for firms to implement strategies for minimising emission risks in industrial processes to address environmental challenges, particularly with regard to CO₂ emissions (Lee and Hussain 2024). Furthermore, by implementing low-carbon operations management in processes and products (Lopes de Sousa Jabbour et al. 2021), firms can design products with recyclability and easier disassembly at the end of a product's life, thereby supporting net-zero transitions (Balci and Ali 2024; Liu et al. 2019).

Low-carbon product material helps in determining and realising the integration of efficient resources needed to design products that incorporate environmental sustainability considerations (Fu et al. 2023; Wanke et al. 2021). The use and reuse of sustainable material are essential for lowering carbon emissions and the hazardous environmental impacts of industrial and supply chain operations (Tapiola et al. 2024). GPI connects the production of goods to minimise energy use, decrease material consumption and reduce GHG emissions throughout the product life cycle (Bag et al. 2022; Chai et al. 2024; Li et al. 2021). Overall, LCOM supports the planning and organisation of production processes that maximise resource efficiency by optimising raw materials and energy using manufacturing processes (Ai et al. 2024; Mangun and Thurston 2002). Thus, we propose the following:

Hypothesis 1. *The adoption of low-carbon product material (H1a) and low-carbon processes (H1b) positively influences a firm's green product development.*

Hypothesis 2. *The adoption of low-carbon product materials and low-carbon processes positively influences closed-loop supply chain practices (H2a) and end-of-product-life management (H2b).*

2.3 | Impact of Green Innovation on the Circular Supply Chain

The environmental benefits of GPI have been widely recognised (Awan et al. 2021). GPI is considered essential to addressing climate change challenges (Chen 2008; Li et al. 2021), such as pollution prevention and resource depletion, as well as meeting customer and market demands while delivering value to society (Awan et al. 2021; Xin et al. 2022). Green products focus on optimising the environmental performance of products through design considerations, such as material selection, energy efficiency and emissions reduction, whereas GPI focuses on developing new innovative products to tackle environmental challenges (Chavez et al. 2023; Chen 2008), integrating low-carbon process technologies (S. Wang et al. 2024) and improving energy efficiency in packaging materials to facilitate end-of-life disposal (Lopes de Sousa Jabbour et al. 2021; Fu et al. 2023).

The integration of resource optimisation, material selection and material efficiency for developing GPI requires firms to optimise resource efficiency and implement technologies (Schiederig et al. 2012). Eco-innovation has been suggested as a process between public policy and the implementation of CSC (Li et al. 2023). Firms with low-carbon operations management can produce products with lower carbon footprints and improved recyclability (Fu et al. 2023; S. Wang et al. 2024), thereby minimising waste, conserving resources and promoting sustainability across the supply chain (Kühl et al. 2022).

The closed-loop concept is closely aligned with CSC management (Mangla et al. 2018). According to Govindan et al. (2015), 'closed-loop supply chain management is the design, control, and operation of a system to maximize value creation over the entire life cycle of a product with dynamic recovery of value from different types and volumes of returns over time' (pp. 603–604). Closed-loop management handles material recyclability, remanufacturing, product recovery and product disassembly to reduce environmental impact (He 2017; Östlin et al. 2008). Reverse logistics includes the return or disposal of products by end users or members of supply networks (Chhimwal et al. 2023; Mishra et al. 2023). Green product end-of-life practices focus on recovering materials or parts from old products so they can be used again (Abdul-Rashid et al. 2017; Fu et al. 2023).

Manufacturing companies can develop and implement operational closed-loop systems to reuse and recover resources throughout the supply chain (Chai et al. 2024; Liu et al. 2019). Circularity in the supply chain enables firms to reduce production costs by promoting reuse and recycling and improving operational efficiency. De Giovanni (2022) asserted that implementing closed-loop and end-of-product-life management requires firms to take into consideration circularity initiatives in the design and production of products with low carbon emissions throughout the product life cycle (Jabbour and de Sousa Jabbour 2014; Mishra et al. 2023).

Hypothesis 3. *GPI positively influences closed-loop supply chain practices (H3a) and end-of-product-life management (H3b).*

2.4 | Moderating Role of Net-Zero Capability

Inefficiencies in end-of-product-life management and closed-loop practices (Chai et al. 2024; Reimann et al. 2019) may result from a lack of LCOM implementation (de Sousa Jabbour et al. 2019). Moreover, implementing GPI depends on the availability of advanced manufacturing processes and technologies within manufacturing processes (Chai et al. 2024; Schiederig et al. 2012). Balci and Ali (2024) emphasised that environmental management capabilities and eco-design competencies enable firms to transition towards net zero. Other research has highlighted that adaptability and responsiveness to the changing environment can significantly improve a firm's ability to integrate GPI (Bag et al. 2022; Li et al. 2021; Liu et al. 2019).

GPI has long been studied in relation to open innovation and external and internal environmental resource management (Awan et al. 2021; Xin et al. 2022). Researchers have operationalised GPI in terms of environmental and carbon performance,

including exploratory green learning and environmental management capabilities (Cui et al. 2023; Song et al. 2020; C. H. Wang and Juo 2021). Dohale et al. (2024) and Bui et al. (2023) emphasised that environmental management-related capabilities (eco-design, sustainable manufacturing and net-zero emissions) reside in the firm's ability to restructure processes and develop internal environmental resource management processes. Balci and Ali (2024) viewed net-zero capabilities as a way to transform decarbonisation processes in the production process, maximise resource productivity and reduce the risks associated with climate change.

Chen et al. (2024) highlighted the critical role of net-zero emissions and carbon neutrality initiatives in enabling firms to develop emission reduction capacities and adopt green practices. According to Dohale et al. (2024), firms with the ability to sense investment opportunities invest in research and development (seizing) to create energy-efficient solutions that reduce emissions, thereby lowering their carbon footprint. Following Teece (2010), a firm's capacity to innovate and integrate new technologies allows it to sense and seize complexities related to sustainability and climate change. These capacities enable firms to develop capabilities to respond to sustainability challenges by integrating cleaner production technologies, investing in process optimisation and improving energy efficiency (Almeida et al. 2013), while maintaining a competitive advantage in the transition to net-zero strategies (Chen et al. 2024). Thus, a firm's DCs attend to the decarbonisation associated with their production processes (Farooque et al. 2022). There are some challenges related to firms' reconfiguring processes, and firms with limited resources may struggle to implement effective net-zero strategies (Kannan et al. 2022). Prior research on net-zero capabilities and sustainable performance showing a positive effect of net-zero capabilities on big data analytics has emphasised that DC is a key driver of sustainable performance (Balci and Ali 2024). From the DC perspective (Teece 2018), net-zero capabilities enable organisations to deal with the uncertain climate challenges of managing resource integration and reconfiguration (Jin et al. 2017; Kannan et al. 2022).

According to Chen et al. (2024), the significance of 'net-zero emissions as an emerging concept in supply chain management is not yet fully clear' (p. 2). A better understanding of net-zero initiatives in supply chain management requires clarity and well-defined strategies to guide firms in their transition to sustainable practices (Chen et al. 2024). In the context of CSC management, net-zero capabilities are important when firms anticipate the benefits of developing and producing products to minimise resource consumption and improve energy efficiency. Thus, we hypothesised the following:

Hypothesis 4. *Net-zero capability moderates the effect of GPI on closed-loop supply chain practices.*

Hypothesis 5. *Net-zero capability moderates the effect of GPI on end-of-product-life management.*

Our conceptual model in Figure 1 explains how low-carbon operations management contributes to manufacturing firms' transition towards CSC management.

3 | Methodology

3.1 | Data Collection

In 2016, Pakistan, a developing nation, launched its decarbonisation strategy in line with United Nations Sustainable Development Goal 12: to encourage the adoption of eco-friendly production technology and facilitate the implementation of eco-standards to enhance low-carbon emissions (Govt. of Pakistan 2021). Manufacturing firms with low-carbon operations (prioritising energy efficiency by optimising production processes) might use environmentally friendly materials and inputs to facilitate product end-of-life disposal (Abdul-Rashid et al. 2017; S. Wang et al. 2024). The Paris Agreement stipulates that emissions must be reduced by 43% by 2030 and reach zero by 2050 (United Nations 2023). Pakistan and other nations joined the Paris Agreement on climate change, which poses risks for developing countries. According to the agreement,

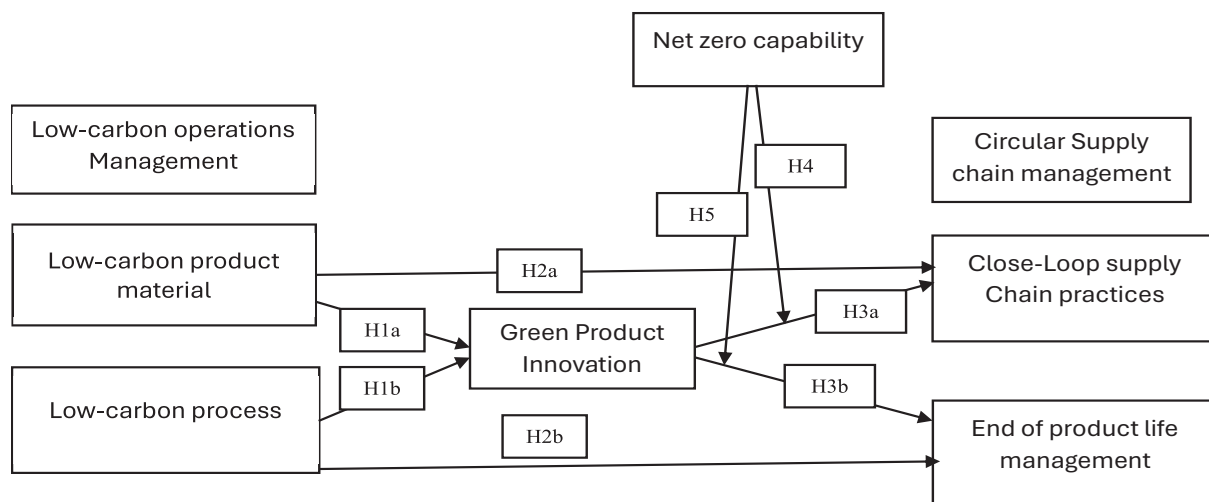


FIGURE 1 | Conceptual framework.

the industry will play a crucial role in assisting Pakistan to achieve its goal of reducing carbon emissions and contributing to global emission reduction efforts. Manufacturing firms are focusing on material reuse and prioritising resource recovery within closed-loop systems to tackle the growing challenge of carbon neutrality in their supply chains (Mangla et al. 2018; Zhang et al. 2023).

To obtain a list of ISO 14001 certified companies, we contacted the accredited certification bodies in Pakistan, which maintain databases of their clients. We assured the accreditation bodies that all shared data would remain confidential and would not be disclosed or shared with any third parties. The study focused on individual companies, and 377 firms were identified with ISO 14001 certification. As a result, 183 survey responses were obtained. The survey questionnaire was in English, as professionals and middle managers in Pakistan frequently speak English. All survey questionnaires were distributed by email from January to July 2024, and follow-up telephone calls were made to increase response rates. We used G*Power analysis software to determine the appropriate sample size (Faul et al. 2009) and ensure that the sample had adequate statistical power ($\alpha = 0.05$). The outcomes confirmed that a sample size of 180 was sufficient to reliably detect meaningful effects and maintain the validity of the study findings (Thomas and Krebs 1997).

The survey questionnaire was created with reference to previous research (Mangla et al. 2018; Zhang et al. 2023). Thirteen experts from the engineering and production departments

assessed it to ensure that the questions were applicable and consistent and to make any necessary revisions.

The experts suggested minor rewording in certain constructs to improve comprehension, and based on their suggestions, adjustments were made to ensure that the questionnaire effectively captured key aspects of low-carbon product material optimisation. The results of the *t*-test showed that the significance value was higher than 0.05, indicating that non-response bias was not an issue. The sample consisted of various managerial roles, including 29 General Managers of Technology (15.8%), 52 Operations/Supply Chain Managers (28.4%), 47 Design/Engineering Managers (25.7%), 35 Directors of Logistics and Distributions (19.1%), and 20 Managers of Production and Purchasing (11.0%). Table 1 provides an overview of the demographic characteristics.

3.2 | Operationalisation of Constructs

We adhered to the suggestions of Flynn et al.'s (1990) by utilising scales and items previously validated in earlier studies as reflecting indicators, as they signify a sample of a substantial population. We included low-carbon process and product operation constructs to assess the extent to which firms implemented low-carbon operations management on a 7-point Likert scale (1 = *not a priority*, 7 = *essential priority*). Low-carbon product material (LCPM) is a set of systematic processes and sustainable practices that organisations use for decarbonisation throughout the product life cycle (Jabbour and de Sousa Jabbour 2014; Lopes de Sousa Jabbour et al. 2021). The scale of Lopes de Sousa Jabbour et al. (2021) was adopted to measure low-carbon operations management. During the pilot testing, respondents emphasised the need to contextualise Gunday et al. (2011) measures as green product innovation. We adopted and conceptualised the GPI scale proposed by Gunday et al. (2011) and measured it on (1 = *not implemented*, 7 = *fully implemented*). We adopted the net-zero capability construct by Balci and Ali (2024) with a 5-item scale to assess levels of agreement with the company's efforts to build capabilities over the past 3 years (1 = *strongly disagree*, 7 = *strongly agree*). To capture respondents' implementation of CSC management, we included closed-loop supply chain practices and end-of-life product management. The scale for end-of-life product management was adopted from the research study by Farooque et al. (2022) and tested in China. We adopted four items from Farooque et al. (2022) to assess the respondents' perceptions of their firms' implementation of end-of-life product management over the past 3 years. For the closed-loop supply chain construct, we used a 6-item scale from De Giovanni's (2022) measure of the construct of closed-loop supply chain practices over the past 3 years (1 = *not at all*, 7 = *to a great extent*). We dropped one item "Offering cheap return options" due to its low factor loading (0.20).

3.3 | Data Analysis

Partial least squares structural equation modelling (PLS-SEM) software (Ringle et al. 2024) was used to examine the structural and measurement models. We used PLS-SEM because

TABLE 1 | Demographic profile of the respondents.

	Frequency
Firm age (years in the business)	
Less than ≤ 10	26 (14.2%)
Between 11 and 20	37 (20.2%)
Between 21 and 30	61 (33.3%)
Greater ≥ 31	59 (32.2%)
Firm size (number of employees)	
< 50	33 (18.0%)
51–250	58 (31.7%)
> 251	92 (50.3%)
Industry categories	
Ceramic and glass industry	29 (15.8%)
Computer and electronic products	33 (18.0%)
Automotive component manufacturing	22 (12.0%)
Electronics component manufacturers	20 (10.9%)
Metal machinery and equipment	14 (7.7%)
Logistics and transportation	32 (17.5%)
Plastics and paper manufacturers	19 (10.4%)
Sports and apparel manufacturers	14 (7.7%)

it is suitable for predictive research (Hair et al. 2020; Sarstedt et al. 2020) and provides good results even in small to medium sample sizes (Dash and Paul 2021). Unlike covariance-based SEM, PLS-SEM does not have strict requirements for data distribution (Hair Jr. et al. 2017).

We conducted a moderation analysis using Hayes and Preacher's (2014) model in SmartPLS software version 4, adding net-zero capability as a moderator for 'closed-loop supply chain' and end-of-product life management. This method provides findings for both high and low levels of moderation, centres variables, and has other benefits for moderation analysis. Since our data came from surveys with Likert scales, which can deviate from multivariate normality, the Hayes and Preacher (2014) method assumes normality of the data and does not require sample splitting, which makes the data appropriate for our research. The descriptive statistics, such as means, standard deviations and correlations, are presented in Tables 2 and 3, which display the factor loadings, variance inflation factors and validity measures. We used SmartPLS 4 software and followed the recommendations of Hult et al. (2018) to assess the discriminant validity of the model. A value of less than 0.90 confirms the reliability of the constructs.

We assessed the construct and discriminant validity of the constructs and tested the internal consistency reliability using Cronbach's α and composite reliability measurements (≥ 0.7) (Hair Jr. et al. 2017). All the constructs' factor loadings exceeded satisfactory levels and were greater than 0.50, and the average variance extracted was greater than 0.50 for each construct, indicating convergence and discriminant validity. As shown in Table 3, the correlation between the latent components for each construct was lower than the square root of the AVE, thereby confirming the discriminant validity of our model (Hair Jr. et al. 2017). To further assess discriminant validity, we utilised Smart PLS 4 software to test the 'heterotrait-monotrait ratio of correlations' (HTMT). HTMT is a superior approach

for assessing discriminant validity, and all our construct values below 0.85 confirmed the reliability of all measures (Henseler et al. 2015).

Further, we followed Hair Jr. et al.'s (2017) recommendations to assess the validity of the structural model using bootstrapping with a resampling number of 10,000 samples, including the significance of path coefficients, and the predictive relevance of the model using cross-validated redundancy, R^2 values, and effect size (f^2). All the endogenous variables' R^2 values (explaining the variance between exogenous and endogenous variables) were within the recommended range.

Finally, following Podsakoff et al. (2003), we conducted Harman's single-factor test in CFA using an unrotated solution with a fixed number of factors. The results showed that no single factor accounted for the majority of the variance, indicating that common method bias is unlikely to be a concern. Collectively, all factors explained 72.07% of the total variance, with the first factor contributing only 19.96%, well below the 50% threshold. In order to address the problem of common method bias, we examined for full collinearity in accordance with the recommendation of Kock and Lynn (2012). By conducting regression analyses concerning a common variable, we assessed whether the variance inflation factor (VIF) was below 5 for each variable (Hair et al. 2020). As our model includes a moderator variable, we assess the strength of the correlation between independent variables, and the VIF values were in the range between 2.7 and 4.9; it suggests that the data do not have any multicollinearity issues (Hair et al. 2020).

3.4 | Structural Model and Moderation Test Results

The full structural model, including the control variables (firm size, company age and number of employees), exhibited good

TABLE 2 | Mean, standard deviation and correlational analysis.

	NZC	LCPd	GPI	EoPLM	LCPr	CLSCP	WE	FS	FA
NZC	1								
LCPd	0.471**	1							
GPI	0.478**	0.405	1						
EoPLM	0.384**	0.301**	0.623**	1					
LCPr	0.050	0.018	0.035	0.122	1				
CLSCP	0.440**	0.332**	0.484**	0.482**	−0.242	1			
WE	−0.134	−0.229**	−0.166*	−0.073	0.079	−0.135	1		
FS	0.040	−0.004	0.002	0.012	−0.138	−0.081	0.207**	1	
FA	−0.118	−0.101	−0.072	−0.090	−0.072	−0.071	0.136		1
Mean	5.1989	4.7177	5.3421	5.2754	5.6922	5.3934	2.688	2.322	2.836
SD	1.1415	1.25398	0.99357	0.94984	0.90411	0.89391	0.8809	0.7630	1.035

Abbreviations: CLSCP: closed-loop supply chain practices; EoPLM: end-of-product-life management; FA: firm age; FS: no of employees; GPI: green product innovation; LCPd: low carbon product; LCPr: low carbon process; NZC: net-zero capability; SD: standard deviation; WS: work experience.

*Correlation is significant at the 0.05 level.

**Correlation is significant at the 0.01 level.

TABLE 3 | Scale and item factors, reliability and validity.

Variable	Item	Factor loadings	AVE	CR	Cronbach's alpha	Collinearity statistics (VIF)
Low-carbon products (LCPd)	LCPd1	0.843	0.758	0.904	0.841	1.834
	LCPd2	0.898				2.121
	LCPd3	0.870				2.076
Low-carbon processes (LCPr)	LCPr1	0.725	0.714	0.881	0.801	1.432
	LCPr2	0.897				2.218
	LCPr3	0.902				2.084
Green product innovation (GPI)	GPI1	0.807	0.623	0.891	0.843	2.722
	GPI2	0.851				3.261
	GPI3	0.837				2.351
	GPI4	0.823				2.179
	GPI5	0.602				1.193
End-of-product-life management (EoPLM)	EoPLM1	0.755	0.675	0.912	0.880	2.258
	EoPLM2	0.788				2.785
	EoPLM3	0.833				1.824
	EoPLM4	0.862				1.708
	EoPLM5	0.864				1.605
Net-zero capability (NZC)	NZC1	0.888	0.747	0.937	0.915	2.927
	NZC2	0.858				2.881
	NZC3	0.875				2.709
	NZC4	0.812				2.035
	NZC5	0.887				3.298
Closed-loop supply chain practices (CLSCP)	CLSCP 1	0.794	0.600	0.882	0.834	1.788
	CLSCP 2	0.855				1.985
	CLSCP 3	0.760				2.050
	CLSCP 4	0.762				2.588
	CLSCP 5	0.691				2.511

predictive relevance and effect sizes. In relation to [H1a](#) and [H1b](#), the low-carbon products were positively related to the GPI ($\beta=0.412$, $t=6.571$, $p<0.05$). The findings showed that when firms implemented low-carbon products, they could put more focus on developing GPI. For [H1b](#), we found a non-significant association between low-carbon processes and GPI ($\beta=0.039$, $t=0.409$, $p>0.05$).

The data did not provide evidence in support of [H2a](#) ($\beta=0.081$, $t=0.0835$, $p>0.05$), and [H2b](#) was also not supported ($\beta=0.11$, $t=1.672$, $p>0.05$). A possible explanation for [H2a](#) not being supported is that low-carbon products are designed to be durable and have long lifespans. In relation to [H3a](#), the analysis revealed that the impact of GPI was significant ($\beta=0.327$, $t=4.04$, $p<0.05$). Our results suggest that the priority of the GPI was to reduce the environmental impact at production. The significant relationship is likely due to firms establishing recycling

processes for new innovative products. For [H3b](#), the impact of GPI on product end-of-life management was positive and significant ($\beta=0.625$, $t=10.36$, $p<0.05$). This suggests that GPI (focusing on efficiency, use of sustainable materials and reduced emissions during production) can ensure responsible disposal methods for products that cannot be reused or recycled.

3.5 | Moderation Analysis

We tested the moderating impact of net-zero capability on the relationship between GPI, closed-loop supply chain and end-of-life product management. Following the literature on moderation (Preacher et al. 2007), a significant interaction effect indicates the presence of moderation. To test the moderation paths described in [H4](#) and [H5](#), we used SmartPLS4 with nonparametric bootstrapping procedures and resampling

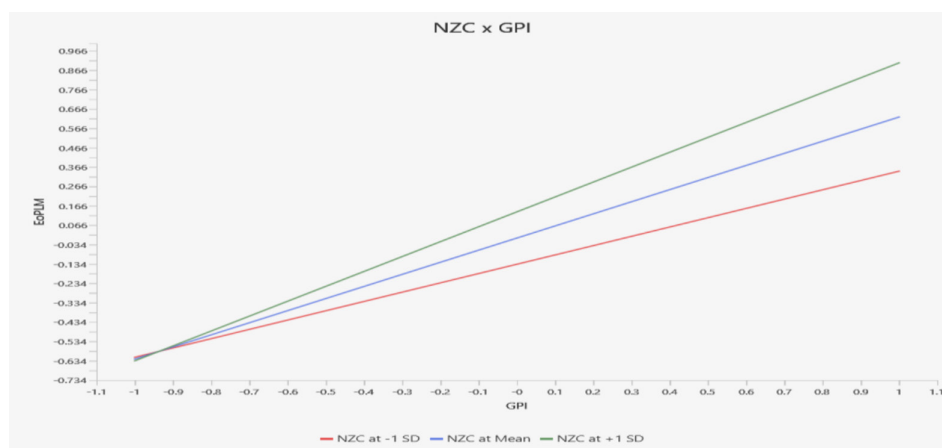


FIGURE 2 | Moderation analysis.

of 10,000 at 95% CI. For H4, we found that the interaction term (net-zero capability $\times$ GPI) not increased the extracted variance for closed-loop supply chain practices (H4) but for end-of-life product management (H5). The path from NZC to CLSCP and EoPLM was positive and significant ($\beta=0.250$, $SE=0.089$, $t=2.804$, $p=0.005$, and $\beta=0.135$, $SE=0.068$, $t=2.003$, $p=0.005$), respectively. The 95% confidence interval (CI) results indicate that net-zero capability did not influence the link between GPI and ‘closed-loop supply chain’ (CLSC) practices ($b=-0.016$, $SE=0.609$, $p=0.795$, $CI=-0.141, 0.092$). This shows that the positive effect of net-zero capability on CLSC practices did not depend on the level of GPI, suggesting that GPI’s effectiveness in advancing CLSC practices may be independent of a firm’s net-zero capability. In other words, firms can successfully implement CLSC practices through GPI, regardless of their net-zero capabilities.

For H5, we found that net-zero capability positively moderated the relationship between GPI and end-of-life product management ($b=0.144$, $SE=0.046$, $CI=0.047, 0.232$). This indicates that net-zero capability helps firms deal with GPI for end-of-life product management. Our results suggest that net-zero capability is a necessary boundary condition for the influence of GPI on end-of-product-life management. Following Aiken (1991), further, we analysed the data at higher (+1 SD) and lower (−1 SD) levels of net-zero capability, and the relationship was positive for high levels of net-zero capability (see Figure 2). The higher the level of net-zero capability, the stronger the positive effect of GPI on end-of-product-life management.

3.6 | Post Hoc Analysis

We conducted an endogeneity analysis using PLS-SEM. Endogeneity is a concern when an independent variable is correlated with error terms, which can arise from reverse causality or omitted variables (Park and Gupta 2012). Following the methodology proposed by Park and Gupta (2012), we evaluated data normality and employed the ‘Gaussian copula’ technique to identify and correct endogeneity biases within our conceptual model. For this purpose, the Gaussian copula method was applied using the maximum likelihood function in the context

of PLS-SEM. The Gaussian copula method streamlines the process of correcting endogeneity biases in regression models, facilitating the attainment of accurate results (Eckert and Hohberger 2023). Its results demonstrate a lack of significant differences between the independent and dependent variables (Park and Gupta 2012). Our robustness checks demonstrated that our model did not consider the error terms present in the dependent variable (Hult et al. 2018).

The omitted variable bias test involves systematically removing one or more variables to check whether the main effects of the remaining variables remain consistent (Chamberlain 1978). Our results indicated that our model provided good stability and reliability for the relationships between variables, ensuring that they were not overly dependent on any single variable. Then, following Small (2007), we used sensitivity to control variable analysis on our conceptual model. We analysed and controlled for firm age, size and number of years in the industry. To further enhance the robustness of our findings, given that our data come from firms in different geographical areas, we incorporated the firm’s location as an additional control variable. The purpose of controlling for firm location was to account for the influence of regional growth and address potential bias. Our findings suggest that the coefficients of the other variables did not change after including the omitted variable, suggesting that the omitted variable was unlikely to be correlated with the dependent variable and did not introduce any bias.

4 | Discussion and Theoretical Contributions

4.1 | Theoretical Contributions

The findings of this study have significant implications for theoretical development, future research and practical application. This study contributes to the CSC management literature by describing the mechanisms through which manufacturing firms achieve circularity in supply chains. Previous studies have conducted empirical investigations of the impact LCOM has on carbon performance (Lopes de Sousa Jabbour et al. 2021; Sheng et al. 2023). This study advances knowledge about the importance of low-carbon operations for

CSC management. Our study is novel and complements previous research by showing that low-carbon operations management can impact carbon emissions (Lopes de Sousa Jabbour et al. 2021; Jabbour and de Sousa Jabbour 2014). Therefore, low-carbon operations management is more likely to facilitate circular production and end-of-life product management (Abdul-Rashid et al. 2017; Fu et al. 2023). Most importantly, our study highlights that firms implementing low-carbon product and process practices may be especially critical for firms needing to improve their CSC management practices. As such, we suggest that low-carbon processes are beneficial for reducing carbon emissions and minimising material waste, essentially creating a circular flow of resources.

This study advances the DC theory literature of Teece et al. (1997), which explains how firms sense, seize and reconfigure their processes and resources to combine and tailor internal and external competencies. Consistent with the DC perspective in supply chain research (Balci and Ali 2024; Lu et al. 2024), we argue that DCs are not only operational in nature but also strategic enablers that allow firms to prioritise specific processes for reducing carbon emissions in products and increase their capability to develop green innovative products. Through resource optimisation and emission reduction in the production process, firms can better develop green product innovations (Bag et al. 2022; Reimann et al. 2019).

Some prior studies have established a direct relationship between GPI and closed-loop and end-of-life product management (Bag et al. 2022; Chai et al. 2024; Liu et al. 2019). Consistent with these previous research studies, this contributes to the theoretical deepening of DCs by considering net-zero capability within the supply chain context, in which firms are better able to develop green product innovation and transition towards achieving end-of-life product management objectives. From a DCs theoretical perspective, firms' net-zero capabilities also contribute to end-of-product-life management, reinforcing the role of net-zero capabilities in CSC management initiatives (Balci and Ali 2024). Following Bag et al. (2022), this study's findings go beyond simply pointing out that GPI is important for implementing end-of-life product management. From a theoretical perspective, net-zero capability extends DC theory by highlighting how firms adapt and reconfigure resources to achieve net-zero goals. It highlights how firms develop, deploy and adapt dynamic capabilities to achieve green innovation and meet end-of-life product management goals.

In contrast to previous research investigations that have concentrated on eco-innovation in CSC (e.g., Li et al. 2023), our research contributes to the CSC management literature by suggesting that GPI is positively associated with end-of-life product management. This study is among the first to examine the moderating role of net-zero capabilities between GPI, closed-loop supply chain practices and end-of-product-life management. The relationship between GPI and end-of-product-life management is contingent on net-zero capability. Our results show that net-zero capabilities moderate the impact of GPI on closed-loop supply chains and end-of-life product management. Firms are likely to develop GPIs that are recyclable, easy to disassemble and manufacturable at high levels of net-zero capabilities. This study provides robust and complementary insights into achieving CSC

management to support net-zero goals (De Angelis et al. 2018; Farooque et al. 2022).

Our findings reinforce the idea that GPI represents a successful mechanism for driving CSC management. However, the effectiveness of GPI increased only at high net-zero capabilities. Net-zero capability is crucial in driving the transition to more effective CSC management by offering innovative green product solutions that minimise waste, improve material recovery and reduce carbon emissions.

4.2 | Practical Implications

The findings of the present study offer several managerial insights for practice. First, our research empirically examines the impact of LCOM on GPI and CSC management. Our results highlight the need for a strategic approach that integrates GPI and end-of-life product management within the framework of low-carbon process operations. Our findings provide valuable insights for managers on how firms can create more sustainable products and effective end-of-product-life management practices, ultimately contributing to their overall net-zero goals (Balci and Ali 2024; United Nations 2023).

Second, our findings show that GPI is linked to closed-loop supply chains and to managing the end of a product's life. For SC managers, our results emphasise the crucial role of GPI in fostering closed-loop supply chain practices and enhancing end-of-product-life management (He 2017; Östlin et al. 2008). They may also prioritise environmentally friendly practices in their production processes to ensure that products have minimal environmental impact throughout their lifecycles (Reimann et al. 2019). Our results highlight that firms need to implement low-carbon operations to play a significant role in the transition towards a low-carbon economy (Morrison et al. 2024).

Third, our results underline that companies must integrate net-zero capabilities with GPI if they are seeking carbon neutrality in a CSC. Net-zero capabilities positively moderate the relationship between GPI and closed-loop supply chain practices. Net-zero capability can play a major role as a boundary-spanning function to accelerate carbon emission reduction (Balci and Ali 2024). This integration allows managers to reduce waste and optimise the use of raw materials, leading to cost savings and more sustainable production processes (Bag et al. 2022; Farooque et al. 2022). We advise managers to invest in building net-zero capabilities to enhance the effectiveness of GPI in managing end-of-life products (Chai et al. 2024; Li et al. 2021).

Based on our finding that a high level of net-zero capability is essential to enhance GPI for end-of-product-life management, production managers can develop resource restoration and recoverability strategies within their supply chains and accelerate their firms' efforts to transition to a low-carbon economy. We recommend that managers consider net-zero capabilities when implementing end-of-product-life management practices. Industrial policymakers should encourage firms to transition towards more effective end-of-product-life management, incorporating low-carbon operations and aiming to achieve net-zero goals.

4.3 | Limitations and Future Research

We collected data from only manufacturing industries in Pakistan, and the research context was constrained to a cross-sectional research design. More studies are needed to strengthen the generalisability in emerging countries, and future research could explore the development of CSC management from a multi-level perspective. This study empirically analysed data using PLS-SEM, but future studies may investigate the integration of FsQCA with additional qualitative and quantitative techniques to identify different contingency factors influencing causal configurations for the implementation of CSC management. Future researchers could use FsQCA with PLS-SEM to investigate what digital capabilities and organisational practice configurations lead to successful low-carbon operations in achieving net-zero goals. As our results showed that net-zero capabilities significantly moderate the relationship, future research on developing net-zero capabilities could include a mediation analysis of organisational resilience and sense-making capabilities.

Comprehensive insights into the specific capabilities needed to implement CSC practices effectively remain lacking, and there is insufficient research on the relationship between GPI and net-zero capabilities. Although net-zero capabilities and GPI are associated with CSC management, there may be other contingent factors that enhance firms' carbon performance. Future studies could examine additional moderating factors and antecedent effects that may impact low-carbon operations and CSC management.

As there is growing recognition of artificial intelligence (AI) in enhancing net-zero performance, future research should explore the development and implementation of AI-enabled net-zero capabilities. We recommend that future researchers explore the impact of internal capabilities on the development of AI-enabled net-zero strategies within the context of supply chains. Data should be collected from different developed countries for cross-comparison to determine the extent to which AI-enabled net-zero capabilities influence firms' circular product innovation.

5 | Conclusions

Our study addresses a critical gap in the literature and provides valuable insights into net-zero capability and green product innovation within CSC management. The purpose was to examine the influence of low-carbon operations on closed-loop supply chains and end-of-life product management, and to empirically investigate the moderating impact of net-zero capability on green product innovation, closed-loop supply chains, and end-of-life product management. The transition towards achieving net-zero goals and reducing carbon emissions remains a critical global priority.

Recently, low-carbon operations management practices have become increasingly important for firms looking to minimise their carbon footprint within their supply chain operations. We conclude that firms adopting GPI with net-zero capability have a stronger capacity to improve end-of-product-life (EPL) management than the closed-loop supply chain. We find that firms that develop and strengthen their net-zero capabilities are more successful in implementing green product innovation to optimise their end-of-life product management. We conclude that,

in the presence of high net-zero capabilities, enables firms to maximise the benefits of GPI initiatives, leading to more sustainable and resource-efficient approaches in handling products at the end of their lifecycle. In the context of operations management, these help us define net-zero capability as follows: net-zero capability refers to a firm's ability to reduce its greenhouse gas emissions and align its operations with decarbonisation efforts.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix 1

Survey Questionnaire

Construct name	Measures and items	Source
Low-carbon product operations	Managers were asked about which low-carbon product operations, processes and logistics practices their organisations have adopted. LCPd 1 'Use of life cycle assessment (carbon footprint)' LCPd 2 'Use of renewable and/or recycled raw materials' LCPd 3 'Reduction of carbon emissions in utilisation phase'	Lopes de Sousa Jabbour et al. (2021)
Low-carbon processes	Managers were asked about which level of adoption, from no adoption to completely adopted (1 = Not a priority, 7 = Essential priority). LCPr 1 'Measurement of carbon emissions in production processes' LCPr 2 'Use of energy/carbon efficient equipment' LCPr 3 'Use of low-carbon/carbon-free energy sources'	Lopes de Sousa Jabbour et al. (2021)
Green product innovation (GPI)	Please indicate the degree to what extent were the product innovations implemented in your organisation in the last 3 years related to the following kinds of activities? (1 = Not implemented, 7 = Fully implemented). GPI 1 'Increasing manufacturing quality in components and reducing emissions in production' GPI 2 'Decreasing manufacturing cost in components and materials for eco-friendly products' GPI 3 'Developing newness for current products leading to improved ease of use for customers' GPI 4 'Developing new products with environmental specifications and functionalities totally differing from the current ones' GPI 5 'Developing new products with components and materials totally differing from the current ones'	Gunday et al. (2011)
Net-zero capability (NZC)	Please indicate your level of agreement with the following statements (1 = Strongly disagree, 7 = Strongly agree). NZC 1 'We collect complete data on carbon emissions from our supply chain members (suppliers and customers)' NZC 2 'We analyse comparable data on the carbon emissions of our supply chain members' NZC 3 'We track the consistency of our supply chain members' carbon emissions' NZC 4 'We transparently monitor the carbon emissions of our supply chain members' NZC 5 'We accurately report the carbon emissions of our supply chain members'	Balci and Ali (2024)
Closed-loop SC practices (CLSCP)	Please indicate your level of agreement with the following statements (1 = Not at all, 7 = To a great extent). CLSCP 1 'Activating closed-loop recycling operations' CLSCP 2 'Investing in refurbishing and remanufacturing of returned items' CLSCP 3 'Improving the information system to track returns' CLSCP 4 'Using specialised third party logistics to acquire returns' CLSCP 5 'Offering replacement goods to consumers'	De Giovanni (2022)
End-of-product-life management (EoPLM)	EoPLM 1 'Collect expired/unsold products from distribution network' EoPLM 2 'Collect used/end-of-life products from customers' EoPLM 3 'Return products to suppliers' EoPLM 4 'Require your main suppliers to collect their packaging materials from your firm (i.e., packaging materials of supplied materials or components)' EoPLM 5 'Collect packaging from customers'	Farooque et al. (2022)